Pattern Recognition and Neural Networks

Assignment 1

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Phase 1: Bio-Optic Sensor Calibration (Dataset 1)

Q1.1 Gradient Descent vs. Closed-Form

Answer: The closed-form solution is: $\mathbf{w}^* = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$

The weight update rule for Batch Gradient Descent is: $\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \cdot \frac{2}{N}\mathbf{X}^\top(\mathbf{X}\mathbf{w}_t - \mathbf{y})$

L2 Norm of Difference: $\|\mathbf{w}_{\text{CF}} - \mathbf{w}_{\text{GD}}\|_2 = \mathbf{0.007001}$

Q1.1 Primal to Dual Translation

Answer: The target variable y was binarized at its median ($= 2.3513$). The hard-margin SVM dual is:

$$\max_{\boldsymbol{\mu}} \sum_i \mu_i - \frac{1}{2}\sum_{i,j} \mu_i\mu_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad 0 \le \mu_i, \ \sum_i \mu_i y_i = 0$$

The given data is not linearly separable. The optimization fails because constraints cannot be satisfied when data is not linearly separable - hence the optimization problem is infeasible.

Phase 2: Oncology Array (Dataset 2)

Q2.3 The Singularity Trap

Answer: Applying the closed-form OLS $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ to Dataset 2 does not give correct result, although python code does not give any error. The number of datapoints ($N = 500$) $<$ the number of features ($D = 1000$), hence the matrix $\mathbf{X}^\top\mathbf{X} \in \mathbb{R}^{1000\times1000}$ has rank at most 500. Therefore $\mathbf{X}^\top\mathbf{X}$ is a singular matrix. This is confirmed by calculating the $\det(\mathbf{X}^\top\mathbf{X}) = 0$, and the condition number $\approx 10^{18}$. When computing the inverse, the matrix becomes extremely ill-conditioned. Because of this, the inversion becomes numerically unstable.

Applying L_2 regularization with $\lambda > 0$, the closed-solution is $(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y}$. Since, $\lambda > 0$ the matrix $(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})$ becomes invertible, hence it gives the correct result.

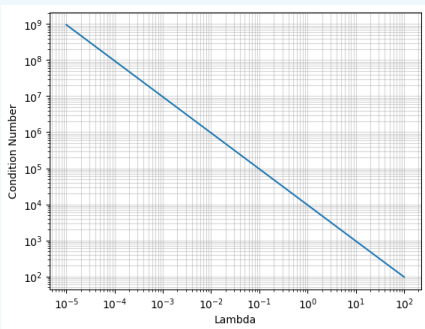
Q2.4 Lasso Sparsity

Answer: In Coordinate Descent, for each epoch, each weight w_j is updated individually using the soft-thresholding operator:

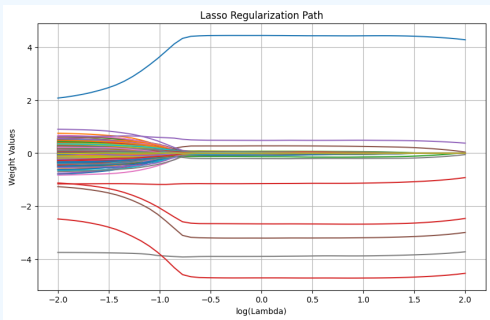
$$w_j \leftarrow \mathcal{S}_\lambda\left(\frac{r_j}{N\|x_j\|^2}\right), \quad r_j = \mathbf{x}_j^\top(\mathbf{y} - \mathbf{X}\mathbf{w} + w_j\mathbf{x}_j)$$

where $\mathcal{S}_\lambda(z) = \text{sign}(z)\max(|z| - \lambda, 0)$.

λ at exactly 50% sparsity: $\lambda^* = \mathbf{0.356}$



(a) Condition number of $(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})$ vs. λ on a log scale. As $\lambda \to 10^2$, the condition number reduces significantly



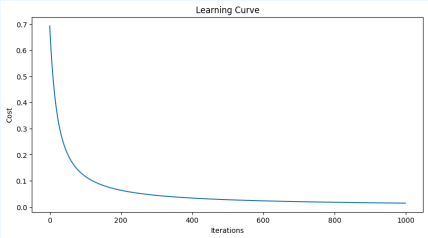
(b) Lasso regularization path: weight values vs. $\log(\lambda)$. As λ increases, the weights progressively converge to zero.

Figure 1: Improvement of condition number in Ridge Regression(left) and Lasso regularization path (right).

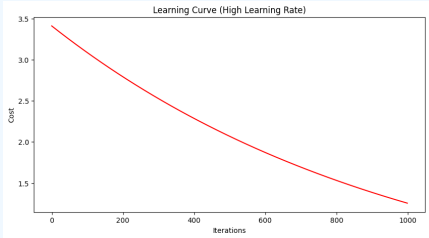
Phase 3: High-Frequency ICU Telemetry (Dataset 3)

Q3.5 Vectorized Logistic Regression

Answer: The gradient of cross-entropy loss with L_2 regularization: $\nabla_{\mathbf{w}}\mathcal{L} = \frac{1}{N}\mathbf{X}^\top(\boldsymbol{\sigma} - \mathbf{y}) + \lambda\mathbf{w}$. The learning rate ($\eta = 50.0$) exceeds $1/L$ where L is the Lipschitz constant. The loss jumps to a very high value at epoch 1 and then decreases very slowly.



(a) Loss curve for $\eta = 0.0001$



(b) Loss curve for $\eta = 40.0$.

Figure 2: Logistic regression loss curves for small η vs. very large η.

Q3.6 GMM Initialization & The E-Step

Answer: A GMM with K = 4 components was initialized with random means and spherical covariances.

Initial Marginal Log-Likelihood: $\mathbf{-2{,}204{,}272.14}$
Log-Likelihood After exactly 1 E-step and 1 M-step: $\mathbf{-1{,}718{,}835.20}$

Q3.7 EM Convergence & Crashes

Answer: Running GMM EM until convergence for $K = 2, 4, 6$ on Dataset 3 empirically confirmed the monotonic convergence property.

Python error when one of the covariance matrices was forced to zero variance mid-training: `numpy.linalg.LinAlgError: When 'allow_singular is False', the input matrix must be symmetric positive definite.`

Forcing a covariance matrix to zero variance makes it singular. The Multivariate Normal PDF requires $\boldsymbol{\Sigma}^{-1}$, due to which the code gives the error.

Q3.8 Decision Boundaries

Answer: A dense grid was constructed over the data range and each grid point assigned to the maximum-posterior class: $\hat{c}(\mathbf{x}) = \arg\max_k \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$.

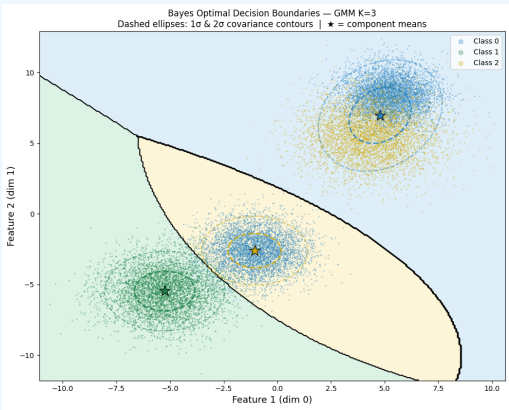


Figure 3: GMM ($K = 3$) Bayes optimal decision boundaries. The contour lines represent the 3σ ellipses of each fitted Gaussian component.

Q3.9 SVM with Slack KKT Verification

Answer: Soft-margin SVM dual is:

$$\max_{\boldsymbol{\mu}} \sum_i \mu_i - \frac{1}{2}\sum_{i,j} \mu_i\mu_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad 0 \le \mu_i \le C, \ \sum_i \mu_i y_i = 0$$

KKT Condition Verification:
(a) **Strictly inside margin ($\mu_i = C$):** No points found
(b) **On margin boundary ($0 < \mu_i < C$):** Index $= 2022$, $\mu = 0.0003$
(c) **Safely outside margin ($\mu_i = 0$):** Index $= 0$, $\mu = 0.0000$

Q3.10 Mercer's Condition

Answer: The RBF (Gaussian) kernel K was computed with $\sigma = 5$. The smallest eigenvalue of K: $\lambda_{\min} = 0.3413$, which is positive. Since all eigenvalues are non-negative, K is Positive Semi-definite (K is, in fact, Positive Definite).

Q3.11 Hyperparameter Topography

Answer:
Most Severe Overfitting Coordinate: $(C, \sigma) = (10,\ 0.001)$
Training Accuracy at this point: 100.0%
Validation Accuracy at this point: 48.0%

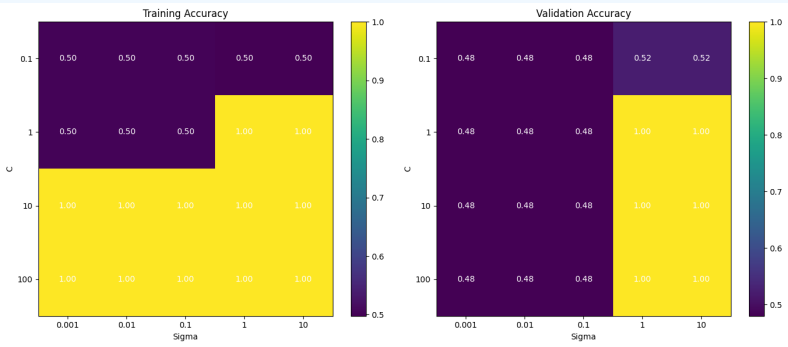
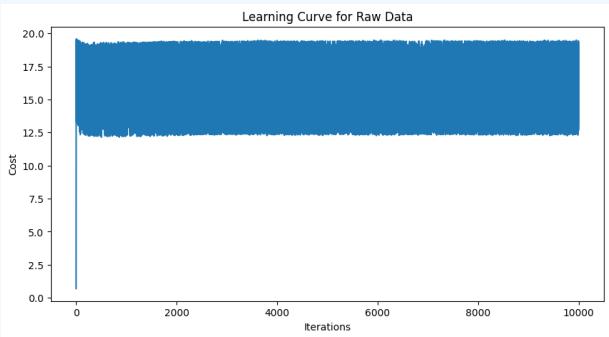


Figure 4: 2D heatmap of validation accuracy across the C-σ grid.

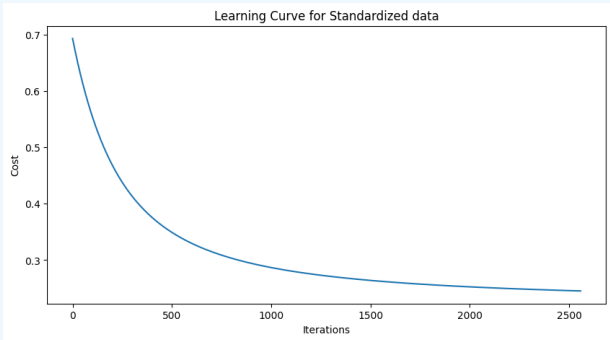
Phase 4: Unscaled Electronic Health Records (Dataset 4)

Q4.12 Multiclass Extension & Scaling

Answer:



(a) Training on raw Dataset 4: model fails to converge within 10,000 epochs.



(b) Training on standardized Dataset 4: loss converges much earlier.

Figure 5: Phase 4.12: Convergence comparison between raw and standardized features.

StandardScaler: $x'_j = (x_j - \mu_j)/\sigma_j$ applied per feature.
Raw Dataset: Model does not converge within 10,000 epochs (the loss keeps oscillating).
Standardized Dataset: Model converges in significantly fewer iterations.

Q4.13 Vectorization Benchmarking

Answer:
Nested for loop Execution Time: 33.437 seconds

Vectorized Execution Time: 1.308 seconds
Speedup Factor: **25.56**×

Q4.14 The Curse of Scale

Answer: Standardization equalizes the contributions of all the features. It allows KNN to use the information of all features rather than being dominated by only the high-scale features.
Validation Accuracy (Raw Dataset): 87.79%
Validation Accuracy (Standardized Dataset): 89.23%

Q4.15 Bayes Optimal & Underflow

Answer: Raw joint probability underflows to 0.0 at $D = 28$. Each Gaussian PDF value becomes < 1 and their product underflows IEEE 754 double-precision.

Phase 5: Bias-Variance Decomposition (Dataset 1)

Q5.16 Empirical Bias-Variance via Bootstrapping

Answer:

Model	Squared Bias	Variance
Linear ($d = 1$)	2.2610	0.001379
Polynomial ($d = 15$)	2.2573	0.009793

Q5.17 Frequentist vs. Bayesian Uncertainty

Answer:

Framework	Expected Value of Slope	Variance of Slope
Frequentist	4.194018	0.000020
Bayesian MAP	4.194084	0.000018

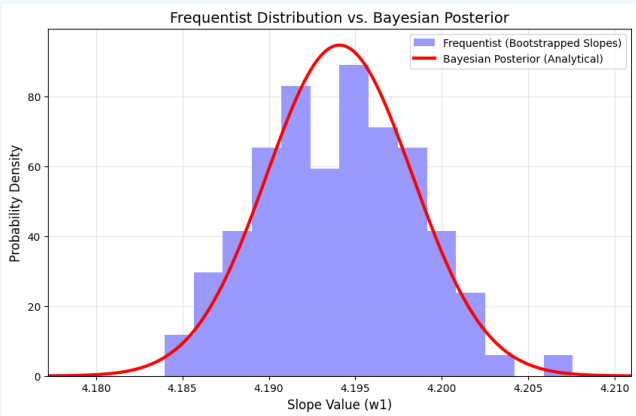


Figure 6: Frequentist vs. Bayesian distributions for the estimated slope parameter w_1.

Frequentist approach: Expectation is over re-draws of the dataset $\mathcal{D}$:

$$\mathbb{E}_{\mathcal{D}}[\hat{w}_1]\,, \qquad \mathrm{Var}_{\mathcal{D}}[\hat{w}_1] = \mathbb{E}_{\mathcal{D}}\Big[(\hat{w}_1 - \mathbb{E}[\hat{w}_1])^2\Big]$$

Bayesian approach: Expectation is over the posterior distributution $p(w_1 \mid \mathcal{D})$:

$$\mathbb{E}_{w_1|\mathcal{D}}[w_1] = \int w_1\, p(w_1 \mid \mathcal{D})\, d\theta, \qquad \Sigma_{\text{post}} = \left(\frac{X^\top X}{\sigma^2} + \frac{I}{\tau^2}\right)^{-1}$$